

3 (a) Meiosis is described as reduction division.

Explain why meiosis is necessary in the life cycle of a sexually reproducing organism.

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(b) Plants need mineral ions to grow and develop. For example, plants need phosphates and a deficiency inhibits cell division and root growth.

Mutations in individuals of some plant populations allow them to survive in mineral-deficient soils.

(i) Name **two** examples of environmental conditions that affect plant phenotype, **other than** mineral deficiency.

For **each** example, describe how it affects the phenotype.

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(ii) Explain why mutations are important in selection.

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(c) A study compared root growth of thale cress, *Arabidopsis thaliana*, in two different soil types:

- full nutrient
- low phosphate.

Two different populations of thale cress were used:

- thale cress with a functional enzyme **X**
- thale cress, with a non-functional enzyme **X**.

30 seedlings from each population were placed in each type of soil and left to grow for seven days. At the start all seedlings had a root of the same length. After seven days, the length of this root was measured again for each seedling.

The mean final root length and standard deviation was calculated for each population of thale cress.

The mean final root lengths are shown in Fig. 3.1.

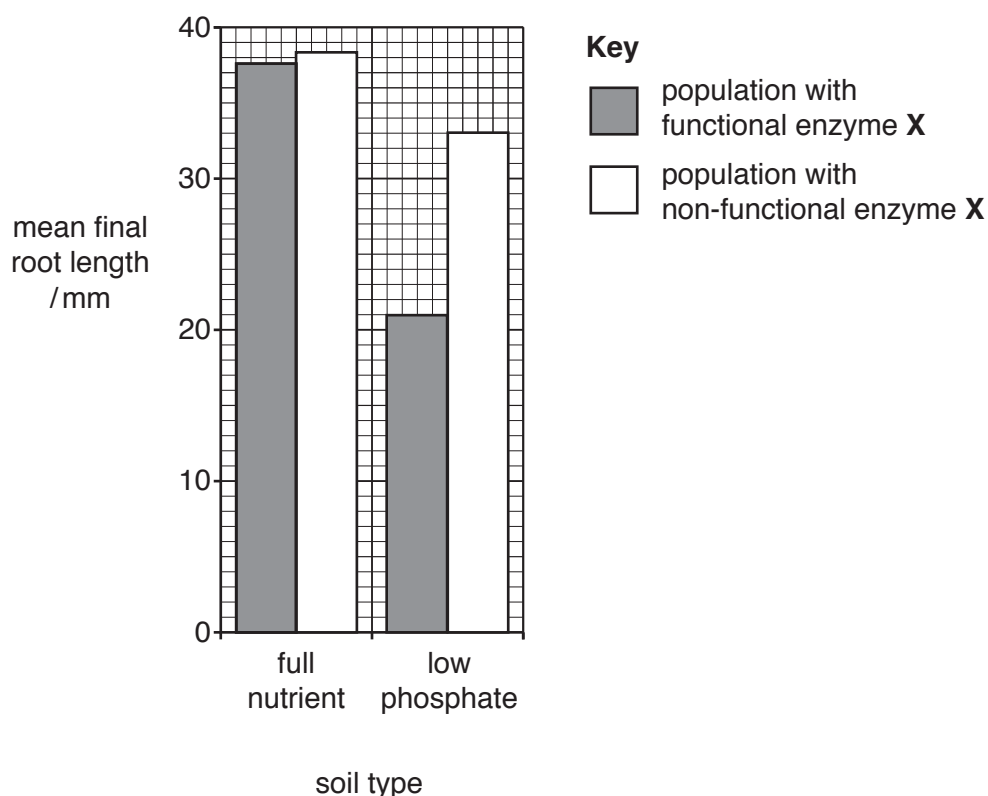


Fig. 3.1

With reference to Fig. 3.1, describe the effect of the low phosphate soil type compared to the full nutrient soil type on root growth, for both populations of thale cress.

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- (d) The null hypothesis states there is no significant difference between the mean final root lengths of the two populations of thale cress grown in low phosphate soil type.

A t -test can be carried out to compare these two means. The critical value for t at the $p=0.05$ significance level is 2.00.

Table 3.1

population grown in low phosphate soil type	mean final root length / mm	standard deviation
functional enzyme X	21	0.5
non-functional enzyme X	33	0.8

- (i) Fig. 3.2 shows the formula for calculating the value of t .

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

$\bar{x}$ = mean
 s = standard deviation
 n = sample size (number of measurements)

Fig. 3.2

Use the formula in Fig. 3.2 to calculate the value of t .

Show your working.

$t =$ [2]

- (ii) Use your calculated value of t to explain whether the null hypothesis should be accepted or rejected.

accept or reject

explanation

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[Total: 13]